

Bale Management Software

Version 1 (Lean): Features and Screen Design

Purpose	One place to see what stock you have, where it is, what it cost to produce, what you sold it for, who owes you, who you owe, and whether the business is making a profit.
Version	Version 1 (lean scope) with photo and bill attachments. Everything else is listed as Version 2.
Status	Draft for customer review. Screens are wireframes showing layout and content, not final visual design. Sample figures are illustrative.

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1. Overview and how the system flows

The system follows the real life of a bale, from purchase of raw material to final payment. Each step feeds the next, so the owner enters data once and sees the whole business on the dashboard.

Step	What happens
1. Purchase	Supplier, quantity, rate, transport and other charges. Raw material stock goes up; supplier balance goes up.
2. Production	Raw material goes into a batch with its own number. Labour and other costs are added. Cost per bale is calculated.
3. Fermentation	Bales are placed in a plot (A1, B7...). The system counts days up to 30 and marks stock ready.
4. Sale	Customer, rate, quantity and invoice. Stock reduces from the exact plot and batch.
5. Payment	Full or part payment. The pending amount becomes customer credit with due date and history.
6. Accounts	Purchases, sales and expenses flow automatically into profit and loss and the dashboard.

Modules in Version 1

Module	What it covers
Purchases	Supplier, material, quantity, rate, extra charges auto-totalled to actual cost, amount paid and pending
Production batches	Batch number, raw material used, labour and other costs, bales made, automatic cost per bale
Plots and fermentation	Each plot shows batch, bale count, day X of 30, ready date and status
Sales and credit	Customer, bales, rate, invoice, amount paid, balance pending, due date, permanent payment history
Expenses	One simple list with category, amount, date, payment mode and note
Owner dashboard	Stock by stage, customer dues, supplier dues, this month's sales, expenses and profit
Basic reports	Customer balances, supplier balances, profit and loss by date range, cost and profit per batch, Excel export
Login with 3 roles	Owner, accountant, staff
Photo and bill attachments	Attach a bill photo or PDF to purchases and expenses; optional photo on sales

2. Design principles, navigation and colour language

- **Simple first.** Every screen has one job. Forms show only the fields needed, and totals calculate on screen as the user types.
- **Phone friendly.** The same web app adapts to a phone. Buttons are large, the camera opens directly for bill photos, and the owner can check numbers from anywhere.
- **Numbers the owner trusts.** Costs, balances and cost per bale are calculated by the system, never typed by hand.
- **Nothing disappears.** Payment records and attachments are never overwritten. Deleting is restricted to the owner and accountant.
- **Indian formats.** Amounts in INR with lakh-style commas, dates as DD-MM-YYYY, units such as kg, ton, load and bale.

Navigation

Device	Layout
Phone	Bottom bar with four tabs: Home (dashboard), Plots, Sales, More (purchases, production, expenses, reports). A large plus button adds a new entry.
Computer / tablet	Left sidebar with the same items plus Purchases, Production, Expenses, Reports and Settings. Top bar shows the user and role.

Colour language (used everywhere)

Amber	Fermenting: bales not yet ready	Green	Ready for sale; also profit and money received
Grey	Empty plot or sold out	Red	Overdue, amount pending, shortfall or loss
Navy	Buttons, headings and totals	Blue	Supplier payable and informational items

Stock statuses

Version 1 uses three statuses for finished bales: **Fermenting**, **Ready for sale** and **Sold**. Raw material is tracked separately as raw material stock. Status changes automatically: a batch becomes Ready when 30 days are complete, and Sold when all its bales are invoiced.

3. Users and permissions

Area	Owner	Accountant	Staff
Dashboard (full, incl. profit)	Yes	Yes	Stock view only
Purchases and supplier balances	Yes	Yes	View quantities only, no prices
Production batches	Yes	Yes	Enter and update; no costs shown
Plots and fermentation	Yes	Yes	Yes (update stock)
Sales, invoices, customer credit	Yes	Yes	No
Expenses	Yes	Yes	No
Reports, profit and loss	Yes	Yes	No
Delete records and attachments	Yes	Yes	No
Add or remove users	Yes	No	No

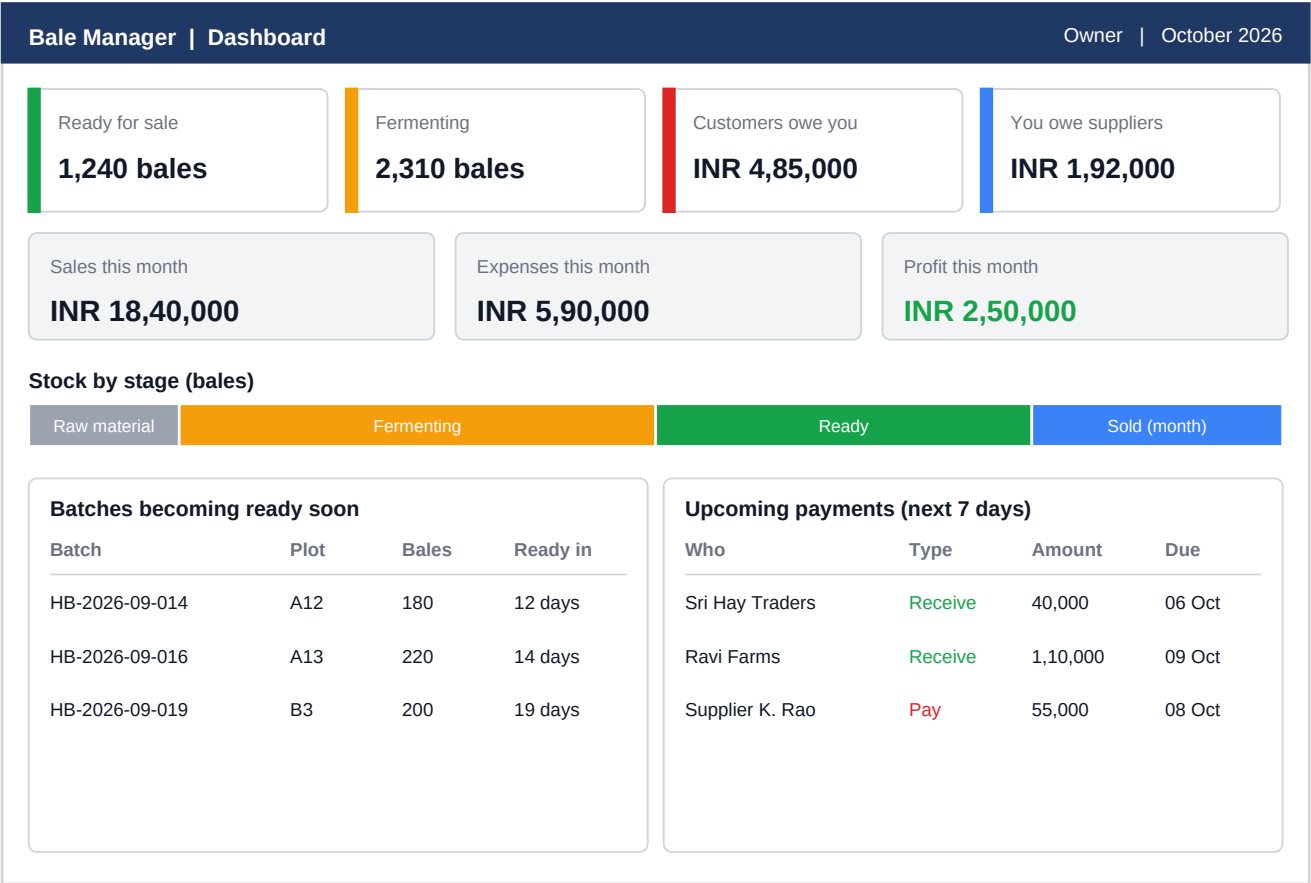
Roles are a proposal. The customer should confirm who sees prices, profit and expenses.

4. Screen designs

Each screen below shows a wireframe of how it should look, followed by the fields and rules. Figures in the wireframes are illustrative.

4.1 Owner dashboard

Purpose: Let the owner understand the whole business within a few seconds of opening the app.



4.2 Purchases and supplier balance

Purpose: Record every raw material purchase with its true total cost, and track what is still owed to each supplier.

Bale Manager | New Purchase

Save | Cancel

Supplier

K. Rao Traders

Purchase date

03-10-2026

Bill / invoice no.

KR-4471

Material

Hay (loose)

Quantity

30

Unit

ton

Rate / unit

4,000

Transport

12,000

Loading / unloading

2,000

Commission

3,000

Other

0

GST rate

0 %

Amount paid now

50,000

Payment mode

UPI

Due date

30-10-2026

Attach bill or photo (image or PDF, max 5 MB)

Take photo

Choose file

bill_1

Save purchase

Auto total

Material

1,20,000

Transport

12,000

Loading

2,000

Commission

3,000

GST

0

Actual cost

1,37,000

Paid

50,000

Pending

87,000

Rate vs last buy

None (first entry)

Wireframe: layout and content only.

Fields and content

- Supplier, purchase date, bill or invoice number
- Material, quantity, unit (kg, ton, load, bale), rate per unit
- Transport, loading and unloading, commission, other charges
- GST rate (single field), amount paid now, payment mode (cash, bank, UPI, cheque), payment due date
- Attach bill or photo from camera or files
- Supplier page: total purchased, total paid, balance pending, list of purchases and payments

Behaviour

- Actual cost = material cost + transport + loading + commission + other + GST. Example: 1,20,000 + 12,000 + 3,000 + 2,000 = 1,37,000
- Pending to supplier = actual cost minus payments made. Later payments add to history and never overwrite it
- Raw material stock increases by the quantity purchased when the entry is saved
- The rate is compared with the previous purchase of the same material and shown as up or down

4.3 Production batch

Purpose: Turn raw material into a numbered batch and know the exact cost of every bale.

Bale Manager | New Production Batch

Save | Cancel

Batch number (auto) HB-2026-10-001	Production date 03-10-2026	Raw material used (kg) 26,500
Raw material cost 1,95,000	Labour charges 18,000	Wrapping / twine 12,000
Machinery / fuel 9,000	Other expenses 4,000	Finished bales 875
Plot assigned A17	Fermentation start 03-10-2026	

Plot is chosen from the A / B / C / D list. Fermentation starts on the date entered.

Save batch

Auto calculation

Total production cost
INR 2,38,000

Cost per bale
INR 272

Raw material / bale
INR 223

Labour / bale
INR 21

Other / bale
INR 28

Ready date
02-11-2026 (day 30)

Figures are illustrative.

Wireframe: layout and content only.

Fields and content

- Batch number (automatic, format HB-YYYY-MM-NNN), production date
- Raw material used (kg) and its cost
- Labour charges, bale wrapping, twine or plastic, machinery and fuel, other expenses
- Number of finished bales produced
- Plot where the bales are placed, and the fermentation start date

Behaviour

- Total production cost = raw material + labour + wrapping + machinery and fuel + other
- Cost per bale = total production cost / finished bales, with raw material, labour and other costs shown per bale
- Raw material stock reduces by the quantity used
- Saving the batch starts the fermentation clock for its plot

4.4 Plots and fermentation

Purpose: Show exactly what stock is in each plot (A1 to A40, B1 to B30, C and D onwards) and how far fermentation has reached.



Wireframe: layout and content only.

Fields and content

- Grid of plot tiles, coloured by status, with bale count and days completed
- Plot detail card: plot, batch number, production date, bales stored, fermentation day X of 30, ready date, status, approximate stock value
- Filter by section (A, B, C, D) and by status; search by plot or batch number
- Move or update stock inside a plot (staff can do this)

Behaviour

- Days completed are counted automatically from the fermentation start date
- Ready date = start date + 30 days (owner can change the number of days in Settings)
- Status switches to Ready for sale at day 30, with no manual step needed
- Approximate stock value = bales in plot x cost per bale of its batch
- Each plot can be named freely, so new plots in C and D can be added without developer work

4.5 Sales, invoice and customer credit

Purpose: Sell bales from ready stock, issue an invoice, and keep a permanent record of every part payment.

Bale Manager | Invoice INV-0042

Customer: Sri Hay Traders

Invoice summary

Invoice total

INR 2,00,000

Amount paid

INR 50,000

Credit pending

INR 1,50,000

Due date

02-11-2026

Bales / rate

700 bales @ INR 285

Stock taken from

Plot	Batch	Bales
A1	HB-2026-08-007	400
A2	HB-2026-08-007	300

Only ready-for-sale plots can be selected.

Stock reduces automatically on save.

Payment history (cannot be edited or deleted)

Date	Mode	Amount paid	Balance after	Note
10-10-2026	UPI	50,000	1,50,000	Advance
25-10-2026	Cash	40,000	1,10,000	Part payment

Add payment

Attach receipt

Pending shows 'Overdue' in red once the due date passes.

Wireframe: layout and content only.

Fields and content

- Customer (name, phone, address, GST number, credit limit)
- Invoice number, date, quantity, rate per bale, total, GST, amount paid, amount pending, due date
- Stock source: plots and batches the bales are taken from (ready stock only)
- Payment history with date, mode, amount, balance after payment and note
- Customer page: total purchased, outstanding amount, last purchase, previous prices, all invoices

Behaviour

- Saving an invoice reduces stock from the chosen plot and batch
- Pending = invoice total minus payments received. Example: 2,00,000 minus 50,000 = 1,50,000; after a 40,000 payment, 1,10,000
- Payment records are append-only. No edit or delete, even for the owner; mistakes are corrected with a new entry
- A simple Overdue flag turns red once the due date passes
- A warning appears if a sale would take the customer over their credit limit

4.6 Expenses

Purpose: Record every business expense in one simple list so profit is always based on real numbers.

Fields

- Paid to, reason, amount, date
- Category (single pick list, see below)
- Payment mode: cash, bank, UPI or cheque
- Related batch (optional)
- Attach bill or photo

Starting categories (editable by the owner)

Category	Examples
Production	Labour, wrapping, twine and plastic, machinery, diesel, electricity
Employees	Salaries, daily wages, overtime, bonuses, advances
Transport	Freight, driver payments, loading and unloading
Factory	Rent, water, repairs and maintenance
Sales	Commission, brokerage, discounts, delivery
Office and admin	Accountant, phone, internet, legal, GST related
Miscellaneous	Anything else

The Version 1 screen is one list with a category pick list; there are no sub-category menus. The list can be filtered by date, category and batch.

4.7 Reports

Purpose: Give the owner and accountant the key reports without extra setup. Every report can be downloaded as an Excel file.

Report	What it shows	Filters
Customer balances	Each customer with total pending and overdue flag	Customer, overdue only
Supplier balances	Each supplier with total purchased, paid and pending	Supplier
Profit and loss	Sales, cost of bales sold, gross profit, expenses by category, net profit	Today, this week, this month, custom range
Cost and profit per batch	Production cost, sales revenue, profit and profit per bale for each batch	Batch, date range

Example profit and loss: Sales 18,40,000 minus cost of goods 12,90,000 = gross profit 5,50,000. Less salaries, electricity, transport and other expenses 3,00,000 = net profit 2,50,000. All values are calculated from entries in the system.

4.8 Photo and bill attachments

Purpose: Keep the proof for every purchase and expense with its record.

- **Where:** purchases and expenses (main use); optionally sales and payment receipts
- **How:** a Take photo button opens the phone camera, and a Choose file button picks an image or PDF
- **Limits:** images and PDFs only, about 5 MB per file; several files per record allowed
- **Viewing:** a small thumbnail on the record opens a full-size view that can be downloaded
- **Control:** staff can add files; only the owner and accountant can delete them
- **Not included:** editing, folders, or automatic reading of bill text (OCR)

4.9 Mobile views

The phone layout uses the same data. It prioritises the dashboard, plot check and quick entry of expenses and bills in the field.

Mobile views (same app, phone layout)

Dashboard

Ready
1,240

Fermenting
2,310

Customers owe
4.85 L

You owe
1.92 L

Profit this month
INR 2,50,000

Ready soon
A12 180 bales 12 days
A13 220 bales 14 days
B3 200 bales 19 days

Home Plots Sales More

Plot A12

Batch
HB-2026-09-014

Bales stored
180

Fermentation
Day 18 of 30

Ready date
15-10-2026

Status
Fermenting

Move / update stock

Home Plots Sales More

Add expense

Paid to
Diesel pump

Amount
3,500

Mode
Cash

Category
Diesel / Electricity

Related batch (optional)
HB-2026-10-001

Photo

File

Save expense

Home Plots Sales More

Wireframe: layout and content only.

5. Calculations the system performs

Item	Formula	Example
Actual purchase cost	Material + transport + loading + commission + other + GST	$1,20,000 + 12,000 + 3,000 + 2,000 = 1,37,000$
Supplier pending	Actual cost minus payments made	$1,37,000 - 50,000 = 87,000$
Total production cost	Raw material + labour + wrapping + machinery and fuel + other	As entered on the batch
Cost per bale	Total production cost / finished bales	$2,38,000 / 875 = 272$
Fermentation day	Today minus fermentation start date	Day 18 of 30
Ready date	Start date + 30 days	03-10-2026 gives 02-11-2026
Plot stock value	Bales in plot x cost per bale of the batch	$180 \times 218 = 39,240$
Customer pending	Invoice total minus payments received	$2,00,000 - 90,000 = 1,10,000$
Gross profit	Sales minus cost of bales sold	$18,40,000 - 12,90,000 = 5,50,000$
Net profit	Gross profit minus all expenses	$5,50,000 - 3,00,000 = 2,50,000$
Profit per bale	Average selling price minus cost per bale	$285 - 218 = 67$

Cost of bales sold uses the cost per bale of the batch each bale came from.

6. Not in Version 1

These features are intentionally left out to keep the first release simple and fast to deliver. They can be added in Version 2.

Feature	Version 1 replacement
E-invoice and e-way bill	Simple GST field and basic GST summary
Multiple GST types and HSN complexity	One tax rate field
Native mobile apps	Mobile-friendly web app
Supplier price comparison and alerts	Price change vs last purchase shown on the purchase screen
Margin analysis beyond per-batch profit	Per-batch profit and profit per bale
Bank and cash flow reports	Not included
Aging buckets (30 / 60 / 90 days)	Simple Overdue flag
Detailed expense sub-categories	Single category list
Six user roles	Three roles
Raw material wastage tally	Add in Version 2

7. Assumptions to confirm with the customer

- Fermentation is a fixed 30 days for every batch (adjustable in Settings), with ready status shown at day 30.
- One batch goes in one plot and one plot holds one batch at a time. Sales are priced per bale; rates may differ by customer.
- Cost of goods sold is calculated from the production cost per bale of the batch sold.
- GST is a single rate per purchase or sale with a basic summary; full GST filing support comes later.
- Permissions in section 3 match how the customer wants staff to see prices and profit.
- Data is cloud-hosted with automatic backups, internet is available at the factory, and the screens are in English.